



Post-Fire Vegetation Recovery in Northern Manitoba **Influence of Burn Severity and Pre-Fire Land Cover**

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Abstract

Vegetation recovery following wildfire was examined in a large 2013 wildfire complex in northern Manitoba, covering approximately 168,837 hectares. This study focused on how burn severity and pre-fire land cover influenced recovery patterns over the first decade after fire. Landsat surface reflectance imagery from 2010 to 2023 was used to calculate the Normalized Difference Vegetation Index (NDVI), the Normalized Burn Ratio (NBR), and the differenced Normalized Burn Ratio (dNBR). These indices were used to assess burn severity and measure post-fire spectral recovery using recovery slope and years to recovery. Annual land cover data were also used to describe broader vegetation change after the fire. Low-severity pixels dominated the burn area, with smaller areas of moderate and very low or unburned classes. Recovery based on NDVI occurred more quickly than recovery based on NBR. Most pixels reached 80% of pre-fire NDVI values within four years, whereas many pixels did not reach the NBR recovery threshold within the study period. Recovery rates differed by burn severity and pre-fire land cover, with moderate-severity areas showing the highest recovery slopes. Land cover results showed that shrub and herb classes remained dominant across the burned area during the first decade after fire. This study provides local insight into post-fire recovery in northern Manitoba and highlights the value of satellite time series for monitoring post-fire spectral recovery in remote boreal landscapes.

Keywords:

Wildfire, Boreal Forest Recovery, Landsat Time Series, Burn Severity, Vegetation Recovery, Northern Manitoba

1.0 Introduction

Fire is a natural disturbance in the boreal forest and has shaped forest composition over long time scales, contributing to the renewal of these ecosystems (Macias Fauria & Johnson, 2008). Lightning-caused wildfires are the primary disturbance in boreal forests and create a mosaic of stands of different ages (Macias Fauria & Johnson, 2008). This relationship is particularly evident in the Boreal Shield ecozone, where fire has shaped forest structure and composition and contributed to renewal over centuries (Frazier et al., 2015). These events not only drive ecological processes but also have social and cultural importance. For many Indigenous peoples, the boreal is a cultural landscape shaped by long-standing relationships with fire (Christianson et al., 2022).

Anthropogenic climate change and colonial forest management practices, including fire suppression, have contributed to ecological and social disruptions associated with wildfires. Some studies link fire suppression to increased fuel loads and changes in wildfire frequency and severity (Mihalus et al., 2024). These ecological changes, together with the broader social impacts of colonial fire governance, have intensified debates over how fire should be governed and who makes decisions about it, particularly on lands used by Indigenous communities

(Zahara, 2020). Understanding long term post-fire dynamics in the Boreal Shield is essential for guiding ecological management and supporting the resilience of northern communities.

Fire records across Canada show that since 1959, both the total area burned and the number of large fires (≥ 200 ha) have increased nationally (Hanes et al., 2019). Over the same period, the national fire season has lengthened starting on average about nine days earlier and ending about seven days later (Hanes et al., 2019). Research in western North America shows that under warmer and drier post-fire conditions, tree regeneration often declines, increasing the likelihood of shifts from forest to non-forest vegetation (Stevens-Rumann et al., 2018). Together, these trends suggest that changing fire regimes may limit post-fire forest recovery (Hanes et al., 2019; Stevens-Rumann et al., 2018), highlighting the need to understand how forests in Manitoba recover after they burn.

There is no single agreed-upon definition of forest recovery, because recovery can be evaluated differently depending on which aspect of the forest is being prioritized. Some studies emphasize structural criteria. For example, using the Food and Agriculture Organization (FAO) definition of forest, a burned area was considered recovered once it had the potential to exceed 10% tree canopy cover and reach a height of 5 m (Bartels et al., 2016). Recovery can also be understood through a cultural lens. For Indigenous communities, a forest may be considered recovered when it again provides important cultural ecosystem services, such as habitat for game or resources that were lost after fire (Christianson et al., 2022). Recovery is often assessed in remote sensing using spectral indicators that track vegetation change over time (Pickell et al., 2016).

Satellite-derived spectral indices from Landsat time series allow post-fire recovery to be monitored across large remote areas (Frazier et al., 2015). They make it possible to examine post-fire recovery over time rather than as a single end point (Frazier et al., 2015). The Normalized Difference Vegetation Index (NDVI) is commonly used to track the first increase in green vegetation after disturbance because it is sensitive to leaf chlorophyll and early regrowth (Pickell et al., 2016). The Normalized Burn Ratio (NBR) is more sensitive to fire-related changes in vegetation structure and condition and is therefore better suited to capturing later stages of post-fire recovery (Pickell et al., 2016).

Burn severity is the degree of change a fire causes to vegetation and soils, especially the loss of organic matter above and below ground (Keeley, 2009). Burn severity is commonly mapped from satellite imagery by comparing pre and post-fire conditions using spectral indices such as the differenced Normalized Burn Ratio (dNBR) (Keeley, 2009; Key & Benson, 2006). This spectral approach makes it possible to map how burn severity varies within a fire over large areas that cannot be measured on the ground (Key & Benson, 2006). This makes burn severity useful for understanding spatial variation in post-fire recovery.

While national and boreal scale analyses exist, there appears to be little published work that focuses specifically on post-fire vegetation recovery within individual large wildfires in Manitoba. In this region, the influence of burn severity and pre-fire land cover on long-term

vegetation change after large wildfires is not well documented. Filling this gap is important for understanding how resilient northern Manitoba's boreal forests are to changing fire regimes and for comparing local recovery patterns to broader trends reported elsewhere. Improved monitoring at this scale can also support northern communities by providing more detailed information on the recovery of lands that are important for cultural uses (Christianson et al., 2022).

To better understand boreal forest resilience, this study analyzes post-fire vegetation dynamics following a large-scale 2013 wildfire event in the northern Boreal Shield. This study addresses three primary questions: (1) How do vegetation greenness and structural recovery evolve over the first decade post-fire? (2) To what extent does burn severity influence the rate of vegetation recovery? (3) How do recovery rates vary across different pre-fire land cover types?

2.0 Study Area

The study area is located in northern Manitoba, Canada, within the Boreal Shield ecozone (Figure 1). On June 6, 2013, a series of lightning-caused wildfires burned a combined area of approximately 168,837 hectares (Canadian National Fire Database, 2021). These fires occurred in the region surrounding the town of Gillam and nearby First Nation and Northern Affairs communities, including Tataskweyak Cree Nation, War Lake First Nation, York Factory First Nation, and Ilford.

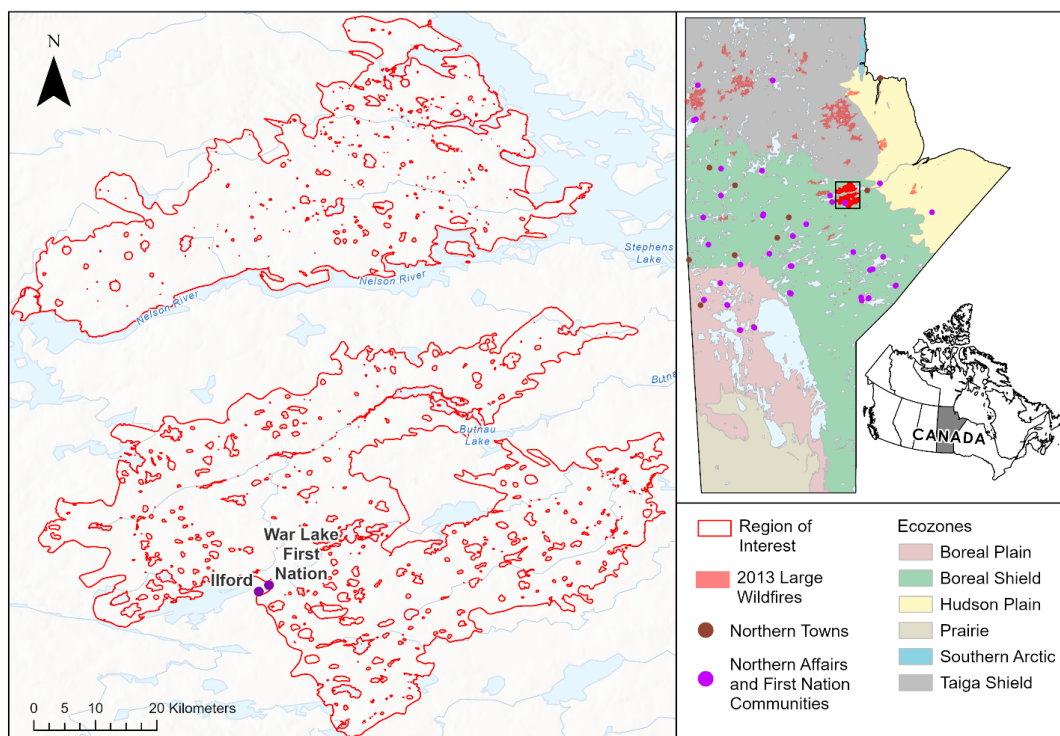


Figure 1. The left figure shows the June 6, 2013 wildfire extent, including the perimeter and burn patterns across a landscape of interconnected lake and river systems. Provincial map of Manitoba (right) showing ecozones, northern towns, northern Affairs and First Nation communities, and large wildfires (greater than or equal to 200 hectares) recorded in 2013, with the study area outlined.

The area lies within the Nelson River watershed, one of the largest drainage systems in the world (Manitoba Government, 2021). The river is part of a connected lake and river system that influences local hydrological conditions. According to the Soils of Canada database (2020), the 2013 burn area lies on organic cryosols, which are carbon rich soils commonly found in northern peatland landscapes and underlain by permafrost.

Based on 1991-2020 climate normal for Gillam A, the average temperature in July is 16.1 °C with average daily maximums of 22.1 °C, while January averages -24.1 °C with average daily maximums of -19.8 °C. Recorded extremes range from 36.4 °C to -46.1 °C (Environment and Climate Change Canada, 2020). The average annual precipitation for the area is approximately 498.4 mm, with the highest monthly totals occurring between June and September (Environment and Climate Change Canada, 2020).

According to the Manitoba Government (2016), forests within the Boreal Shield ecozone are dominated by coniferous species such as black spruce, jack pine, white spruce, tamarack, and balsam fir. The most common deciduous species are trembling aspen, white birch, and poplar. Based on the 2012 species composition layer (Hermosilla et al., 2024), black spruce was the dominant tree species in the area prior to the 2013 wildfires.

3.0 Data

3.1 Remote sensing imagery

Landsat imagery is the primary dataset used to evaluate burn severity and recovery. All imagery was downloaded from the USGS EarthExplorer portal as Landsat Collection 2 Level-2 Surface Reflectance (SR) products, using the Landsat 4-5 TM C2 L2 and Landsat 8-9 OLI/TIRS C2 L2 datasets. The Landsat images of interest to this study are found at Path 032 Row 21 according to the Worldwide Reference System (WRS).

Landsat 5 TM SR imagery from 2010 and 2011 is used to describe pre-fire conditions. Landsat 7 ETM+ scenes from 2012 are excluded because the scan line corrector failure causes data gaps across each image. Post-fire conditions are described using Landsat 8 OLI SR imagery from 2013-2023, which has comparable spectral bands and 30m spatial resolution, allowing conditions before and after the fire to be compared. For each year in this period, all Landsat 5 and Landsat 8 SR scenes with less than 70% scene-level cloud cover and acquisition dates between 1 June and 31 August are included in the dataset.

3.2 Vegetation data

Vegetation types are described using land cover products from National Terrestrial Ecosystem Monitoring System (NTEMS). NTEMS provides annual land cover maps for Canada at 30 m resolution derived from Landsat time series (Hermosilla et al., 2018). NTEMS land cover classes are treated as vegetation types and used to describe pre-fire vegetation composition within the 2013 fire perimeter.

3.3 Fire perimeter data

Fire perimeter boundaries were obtained from the Canadian Wildland Fire Information System (Canadian National Fire Database, 2021). The study focuses on three 2013 fires with ID values NE063, NE064, and NE065. These polygons were selected and merged into a single wildfire complex perimeter. The resulting perimeter defines the study area and is used to define Landsat and NTEMS data.

4.0 Methods

This section describes the steps used to analyze burn severity, vegetation recovery, and post-fire land cover dynamics within the June 6, 2013 fire perimeter. The workflow is summarized in Figure 2. All analyses were raster based and completed in R.

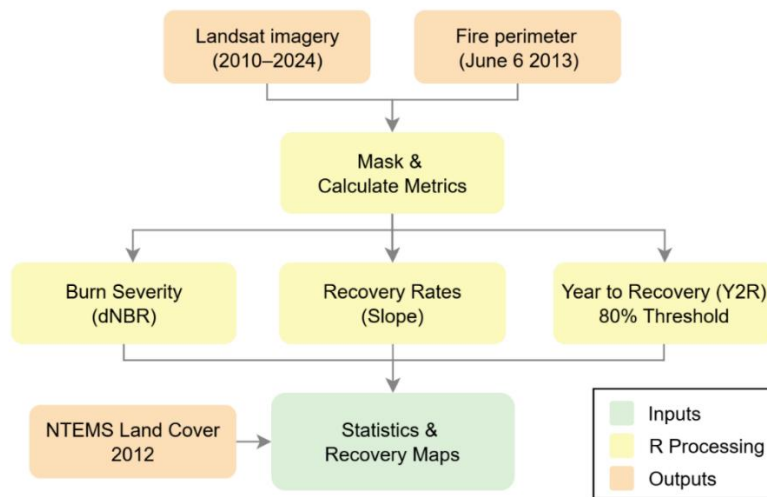


Figure 2. Workflow used to analyze burn severity, vegetation recovery, and land cover dynamics within the June 6, 2013 fire perimeter.

4.1 Data preparation and spectral index calculation

Landsat imagery was clipped to the fire perimeter to ensure the same spatial extent. To reduce contamination from clouds and other poor quality observations pixels were masked using the Landsat QA layer and only clear pixels were kept for analysis. The red, near-infrared (NIR), and shortwave infrared (SWIR2) surface reflectance bands were extracted and used to calculate two spectral indices: the Normalized Difference Vegetation Index (NDVI) and the Normalized Burn Ratio (NBR). NDVI and NBR were derived using the following equations:

$$\text{NBR} = \frac{\text{NIR} - \text{SWIR2}}{\text{NIR} + \text{SWIR2}} \quad \text{NDVI} = \frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}}$$

NDVI was used to describe changes in vegetation greenness through time. NBR was used to describe post-fire spectral change associated with fire effects. Annual NDVI and NBR rasters were created using median compositing of all clear observations within each year to produce one NDVI and NBR layer per year. These annual median composites were then stacked into multi-

year NDVI and NBR time-series datasets to support pixel-based burn severity and recovery analyses in subsequent steps.

4.2 Burn severity

Burn severity was quantified using the differenced Normalized Burn Ratio (dNBR), calculated as the difference between pre-fire and post-fire NBR:

$$\text{dNBR} = \text{NBR}_{\text{pre}} - \text{NBR}_{\text{post}}$$

Pre-fire conditions were represented using 2011 Landsat imagery. Post-fire conditions were represented using 2014 imagery, as scenes acquired in 2013 were affected by smoke and cloud contamination within the fire perimeter. dNBR values were classified into burn severity classes using thresholds adapted from the FIREMON Landscape Assessment method (Key & Benson, 2006) (Table 1). These thresholds represent increasing levels of spectral change associated with fire effects.

Table 1. Burn severity classification levels and dNBR thresholds adapted from Key and Benson (2006).

Pixel Value	Severity Level
< 0.100	Very Low / Unburned
0.100 to 0.269	Low
0.270 to 0.439	Moderate
> 0.440	High

4.3 Spectral recovery metrics

Post-fire vegetation recovery was quantified using two metrics derived from annual NDVI and NBR time series: (1) recovery rate (slope) and (2) years to recovery (Y2R).

4.3.1. Recovery rate

Recovery rate was estimated by fitting a linear regression to annual NDVI and NBR values from 2014 to 2023 for each pixel:

$$\text{Index}_t = m \times t + b$$

where t represents years since fire (with $t = 1$ in 2014 and $t = 10$ in 2023), and m represents the slope of the fitted line. The slope coefficient was used as the recovery rate. Slope values indicate the direction of change in spectral values through time. Positive slopes indicate increasing values; near-zero or negative slopes indicate little change or declining values.

4.3.2 Years to recovery

Y2R was used to estimate the timing of spectral return towards pre-fire conditions. For each pixel a baseline was calculated as the mean NDVI and NBR value from 2010 and 2011. A recovery threshold was defined as 80% of this baseline value. Y2R was defined as the first post-fire year in which the spectral index met or exceeded the recovery threshold for two consecutive years. The recovery year was recorded as the first year of that consecutive pair. Pixels that did

not meet this threshold within 10 years following the fire were classified as not recovered within the analysis period.

4.4 Statistical analysis of recovery drivers

Recovery slopes were analyzed using a two-way ANOVA. This tested for differences among burn severity classes and pre-fire land cover classes and whether the effect of burn severity on recovery varied by land cover type. Burn severity class (from dNBR) and pre-fire land cover class (from NTEMS) were included as factors, along with their interaction term. This analysis was conducted separately for pixel based NDVI and NBR slopes.

Prior to analysis the dataset was limited to vegetated land cover classes (Coniferous, Mixedwood, Shrubs, Wetland, Wetland-Treed). To ensure balanced sample sizes a random sample of 1000 pixels was taken from each group defined by land cover type and burn severity. A fixed random seed was used to maintain reproducibility. Where significant effects were found, Tukey's HSD post-hoc tests were used for pairwise comparisons. Recovery slopes were expected to differ among burn severity classes and pre-fire land cover classes, with the influence of burn severity varying across vegetation types.

4.5 NTEMS land cover composition

Annual NTEMS land cover classifications (2011-2022) were used to summarize land cover change within the fire perimeter. Each yearly NTEMS layer was clipped to the study area and re-indexed by years since fire (-2 to +9) relative to the 2013 fire year. For each year the percentage of total area occupied by each NTEMS land cover class was calculated. These yearly proportions were visualized using a stacked area plot to show shifts in land cover composition before and after the fire. This shows how the proportion of the fire perimeter occupied by each land cover class changed across the recovery period and allows comparison with broader NTEMS land cover patterns reported by (Hermosilla et al., 2018).

5.0 Results

5.1 Burn severity distribution

The burn severity map (Figure 3) illustrates the spatial distribution of burn severity classes across the burn area. Low severity accounts for the majority of the mapped burn area. Moderate severity is distributed in more localized patches. Very low/unburned areas were concentrated along water bodies and also occurred as smaller patches throughout the fire perimeter. Very low/unburned patches were visible in the eastern part of the northern portion of the burn extent and along the southwestern edge of the southern burn extent (Figure 3). The distribution of dNBR values within the fire perimeter (Figure 4) was dominated by the low severity class, with smaller portions classified as moderate and very low/unburned. No pixels were classified as high severity using the selected thresholds.

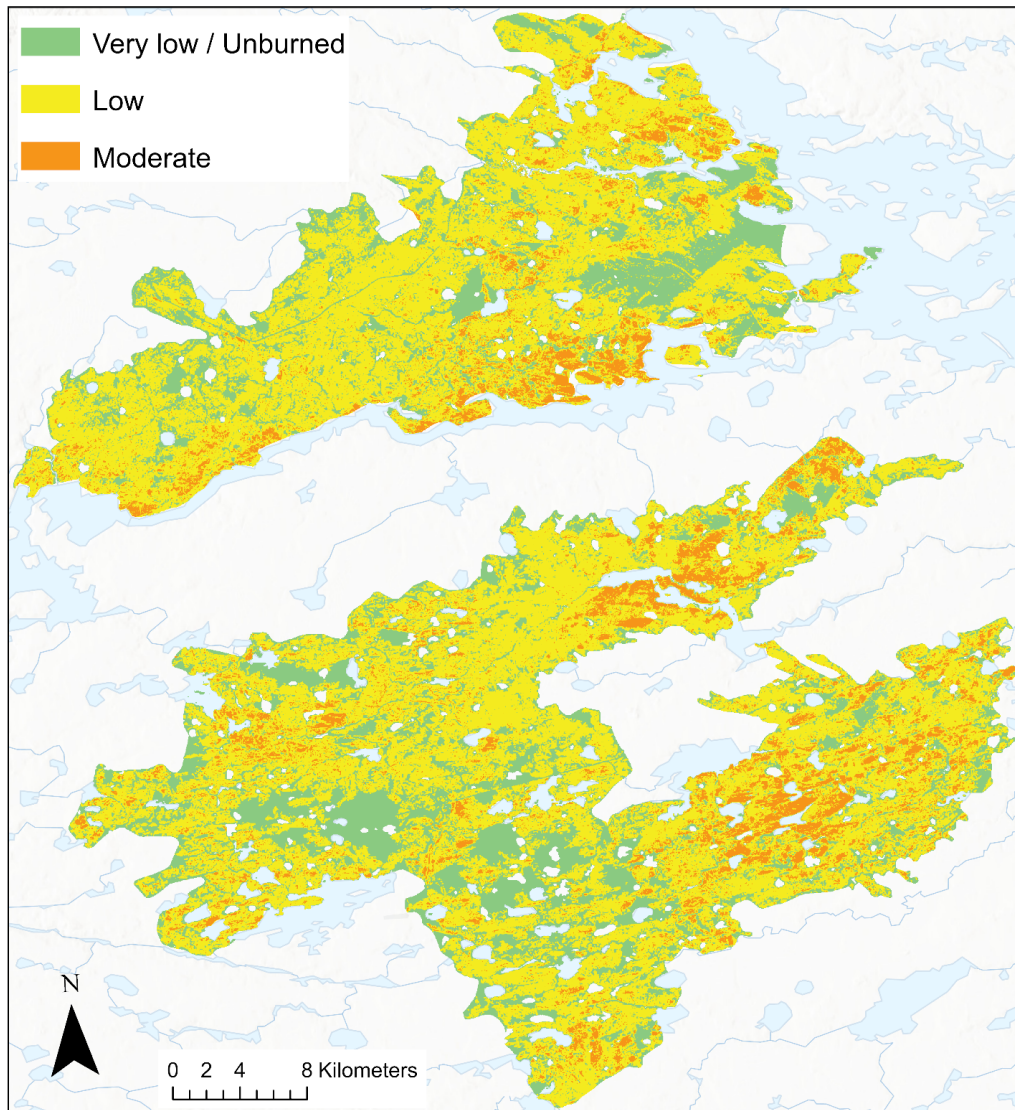


Figure 3. Spatial distribution of pixel-based burn severity classes based on the differenced Normalized Burn Ratio (dNBR) within the fire perimeter. Pixels were classified using the selected thresholds (very low/unburned <0.100 ; low $0.100-0.269$; moderate $0.270-0.439$) to map the pattern of burn severity across the burn extent.

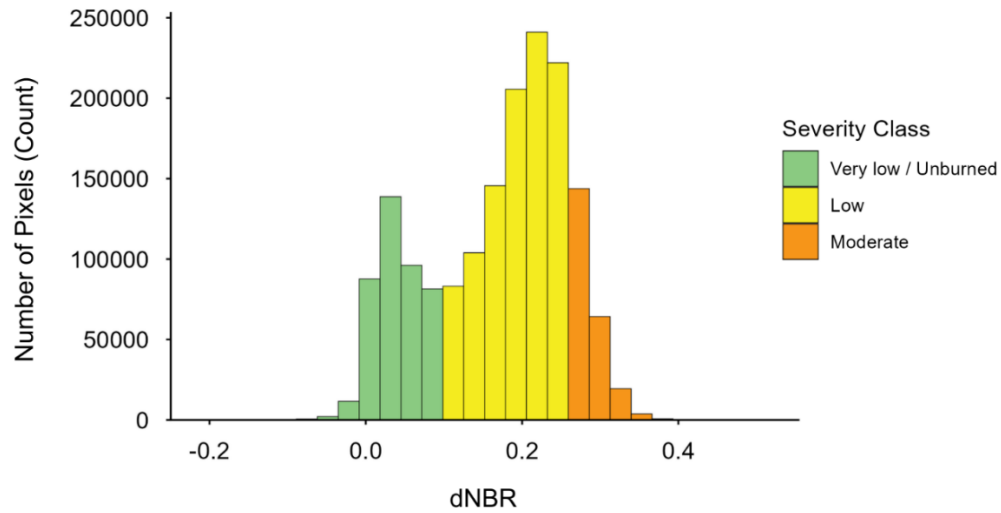


Figure 4. Distribution of pixel level burn severity values based on the differenced Normalized Burn Ratio (dNBR) within the fire perimeter. Bars show the number of pixels in each dNBR interval, grouped by severity class: very low/unburned (<0.100), low ($0.100-0.269$), and moderate ($0.270-0.439$).

5.2 Spectral recovery patterns

5.2.1 Recovery rate

Spatial patterns of post-fire recovery slope for NDVI and NBR are shown in Figure 5. Positive slope values dominate across the fire perimeter for NDVI and NBR, suggesting increasing spectral values from 2014 to 2023. NDVI slopes (Figure 5A) show relatively consistent positive recovery across both burn extents. NBR slopes (Figure 5B) exhibit a similar overall pattern but extend across a broader range of slope values.

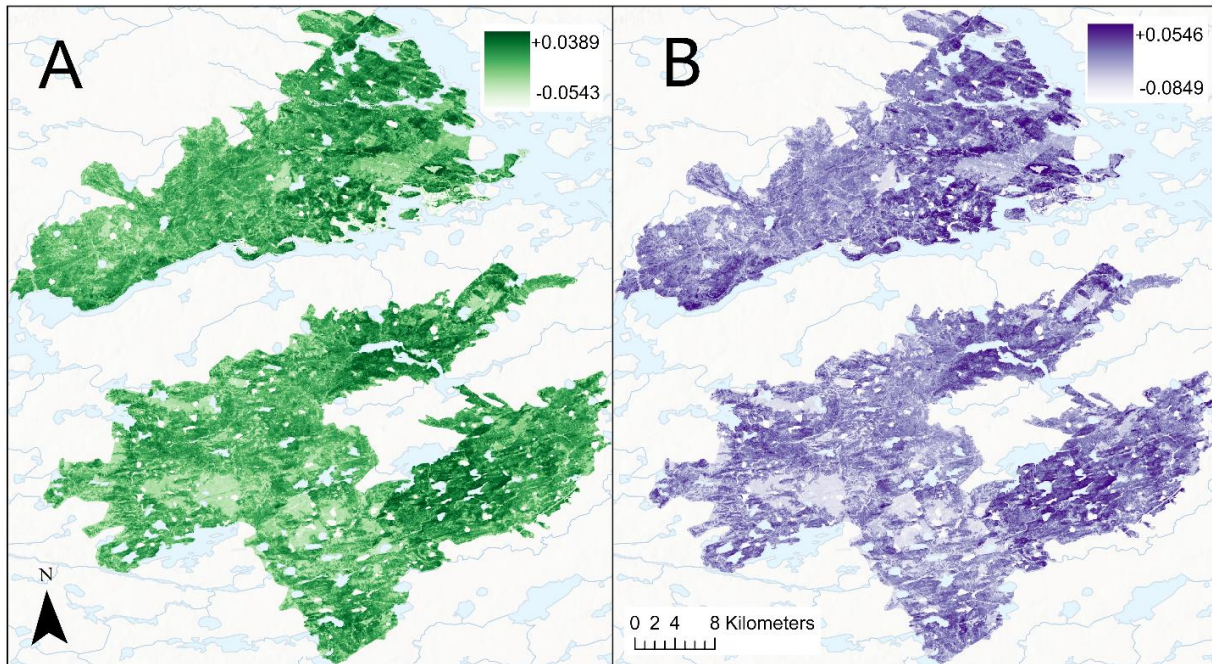


Figure 5. Spatial distribution of post-fire spectral recovery rate (2014-2023) within the fire perimeter. (A) NDVI recovery slope and (B) NBR recovery slope calculated for each pixel using linear regression of annual spectral values from 2014 to 2023. Positive values (dark green) indicate increasing spectral response through time, negative values indicate declining or limited recovery.

5.2.2 Years to recovery

The Y2R analysis identified a difference in the temporal and spatial trends between the two spectral indices. Figure 6 presents the distribution of recovery years. NDVI reaches the 80% recovery threshold earlier than NBR. NDVI recovery was concentrated within the first four years following the fire with the largest proportion of pixels reaching recovery by year 3. As shown in Figure 7A, this recovery of vegetation greenness is visible across the northern and southern fire perimeters.

The NBR recovery timeline varied, with recovery years spread across the entire 10-year study window. A large proportion of the study area remained in the not recovered (NR) category by 2023. As shown in Figure 7B, NBR recovery displays a "salt and pepper" pattern of scattered recovery throughout the burn area. However, larger patches of non-recovered pixels are concentrated in the eastern portion of the northern fire and across the southern fire extents. Pixels reaching the recovery threshold by 2014 match the locations of the very low/unburned severity class within the fire perimeters.

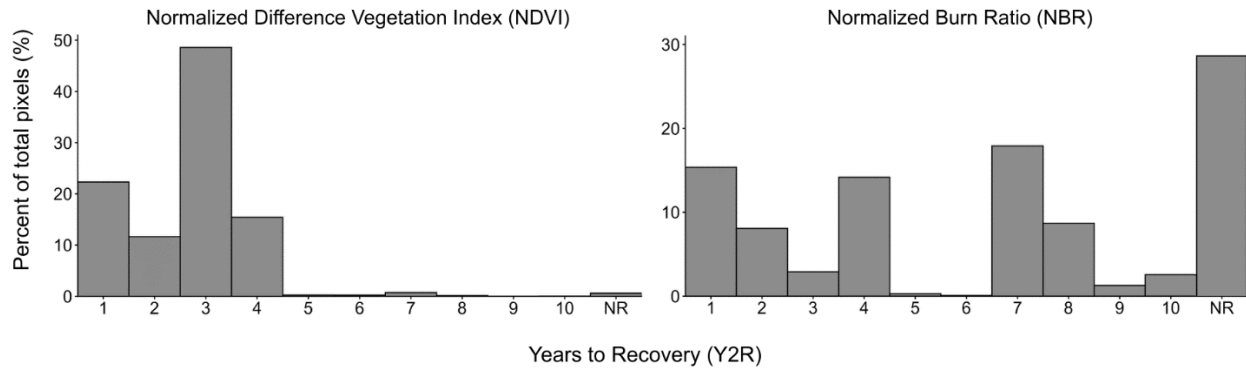


Figure 6. Distribution of Years to Recovery (Y2R) derived from the Normalized Difference Vegetation Index (NDVI) (left) and the Normalized Burn Ratio (NBR) (right) within the fire perimeter. Bars show the percentage of total pixels assigned to each recovery year category (1-10 years). Pixels classified as NR (not recovered) represent areas that did not reach the recovery threshold within the analysis period.

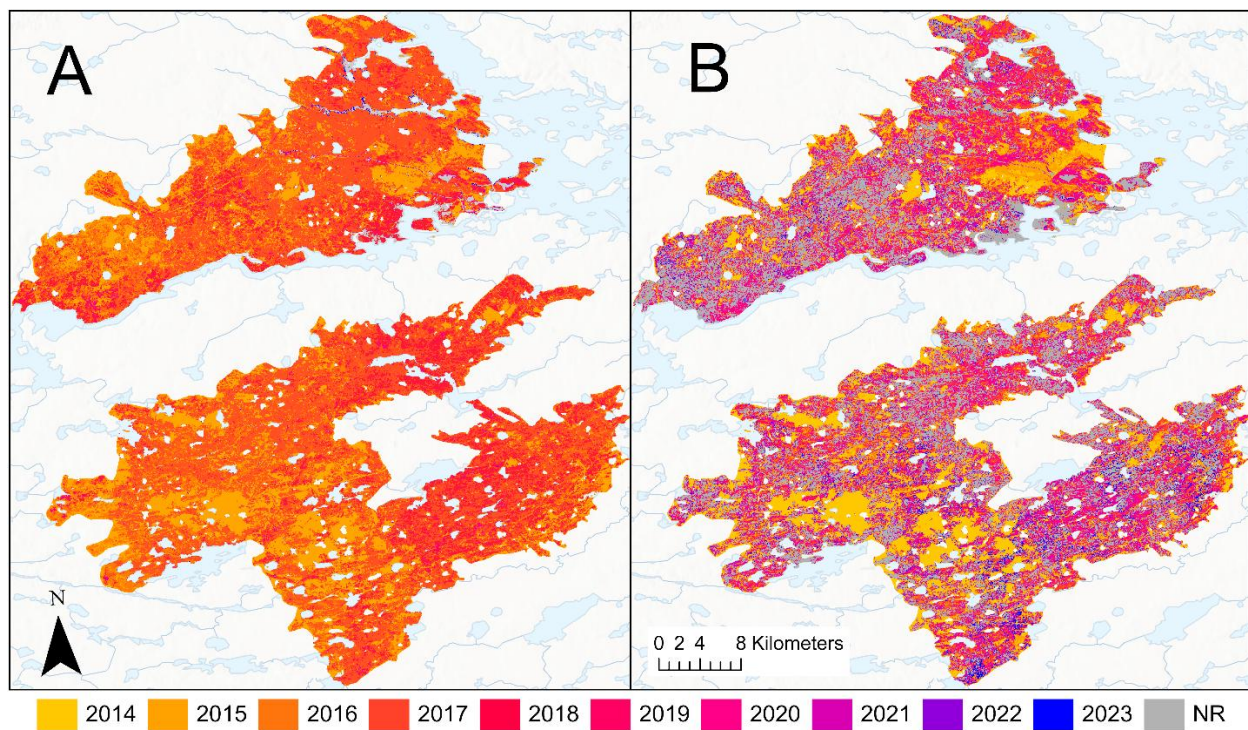


Figure 7. Spatial distribution of Years to Recovery (Y2R) within the fire perimeter. (A) NDVI and (B) NBR showing the first post-fire year in which each pixel reached 80% of its pre-fire baseline value for two consecutive years. Colours represent recovery year (2014-2023) and NR (not recovered) indicates pixels that did not meet the recovery threshold within the 10-year analysis period.

5.3 Effects of burn severity and land cover on recovery

Burn severity was significantly associated with recovery slopes for both spectral indices. Severity class was significant for NDVI slopes ($F = 12,286$, $p < 0.001$) and NBR slopes ($F = 18,207$, $p < 0.001$). Tukey's HSD comparisons indicated that recovery slopes differed among all

severity classes. As shown in Figure 8, mean recovery slopes increased from very low/unburned to moderate severity for NDVI and NBR.

Land cover class was also significant for NDVI ($F = 51.46$, $p < 0.001$) and NBR ($F = 139.5$, $p < 0.001$). Shrub dominated areas exhibited the highest mean recovery slopes, while wetland and wetland-treed classes showed the lowest slopes. Although statistically significant the differences among land cover classes were smaller in magnitude than those observed among burn severity classes. Table 2 provides context for severity patterns by summarizing the minimum, maximum, and range of dNBR values within each land cover class.

The interaction between burn severity and land cover was significant for both NDVI ($F = 31.19$, $p < 0.001$) and NBR ($F = 34.52$, $p < 0.001$), indicating that the magnitude of the severity effect varied across land cover types. In all land cover classes, the recovery slopes increased from very low/unburned to moderate severity.

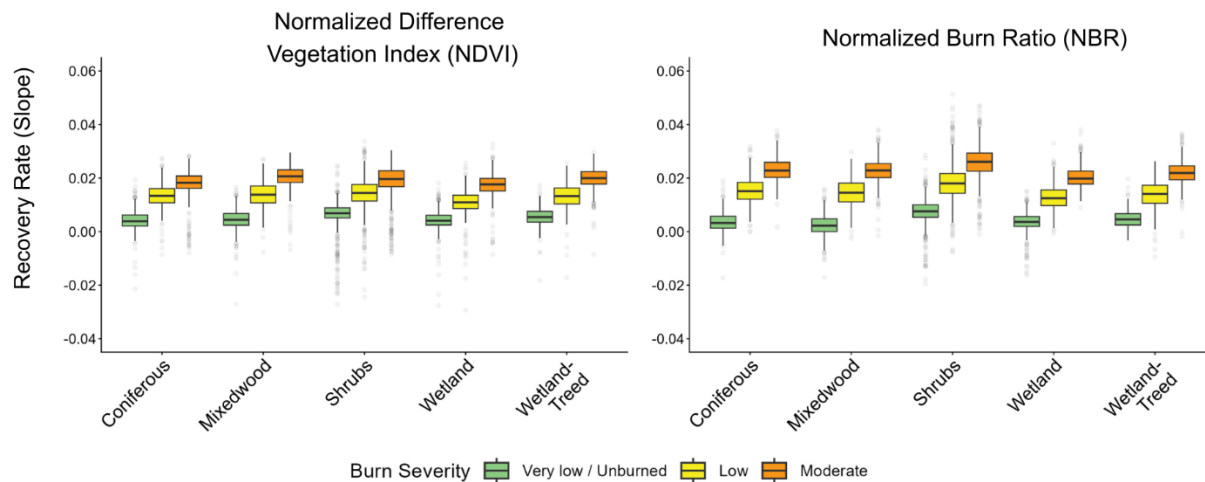


Figure 8. Recovery rates (slope of spectral index change through time) summarized by pre-fire land cover class and burn severity. Boxplots show recovery slopes for the Normalized Difference Vegetation Index (NDVI) (left) and the Normalized Burn Ratio (NBR) (right) across five land cover classes (coniferous, mixedwood, shrubs, wetland, and wetland-treed). Colours indicate burn severity classes derived from the differenced Normalized Burn Ratio (dNBR): very low/unburned, low, and moderate severity. Points represent individual pixel values and boxes summarize the median and interquartile range of recovery slopes.

Table 2. Burn Severity (dNBR values) across the major land cover classes found in the wildfire area.

Land Cover Type	Minimum	Maximum	Range
Shrubs	-0.10	0.40	0.50
Wetland	-0.24	0.42	0.65
Wetland-Treed	-0.07	0.40	0.47
Coniferous	-0.15	0.40	0.55
Mixedwood	-0.08	0.43	0.51

5.4 Land cover composition through time

Changes in NTEMS land cover composition within the fire perimeter are shown in Figure 9. In the two years before the fire (2011-2012), the area was dominated by coniferous forest, with wetland and wetland-treed classes also making up a noticeable portion of the landscape. In the fire year (2013) coniferous cover dropped sharply and the perimeter shifted to mostly shrub and herb classes. Herbs made up a large proportion of the mapped area in the first few post-fire years (2014-2016) then declined by around year 4. Shrubs became the dominant class across the remaining post-fire period. Coniferous cover remained much lower than pre-fire conditions through 2022.

These compositional changes are also reflected spatially in the annual land cover maps (Figure 10). The maps show the shift from large pre-fire coniferous areas (2011-2012) to widespread shrub and herb cover beginning in 2013. Wetland and wetland-treed classes remained in similar locations across years.

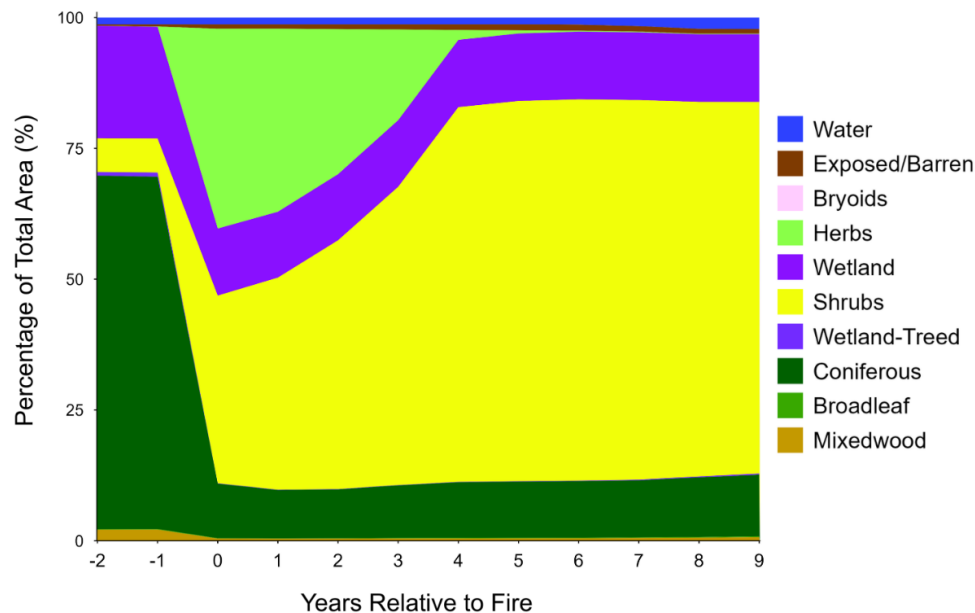


Figure 9. Land cover composition through time within the fire perimeter based on annual classifications from the National Terrestrial Ecosystem Monitoring System (NTEMS). The stacked area plot shows the percentage of total area occupied by each land cover class for each year relative to the 2013 fire year (-2 to +9 years).

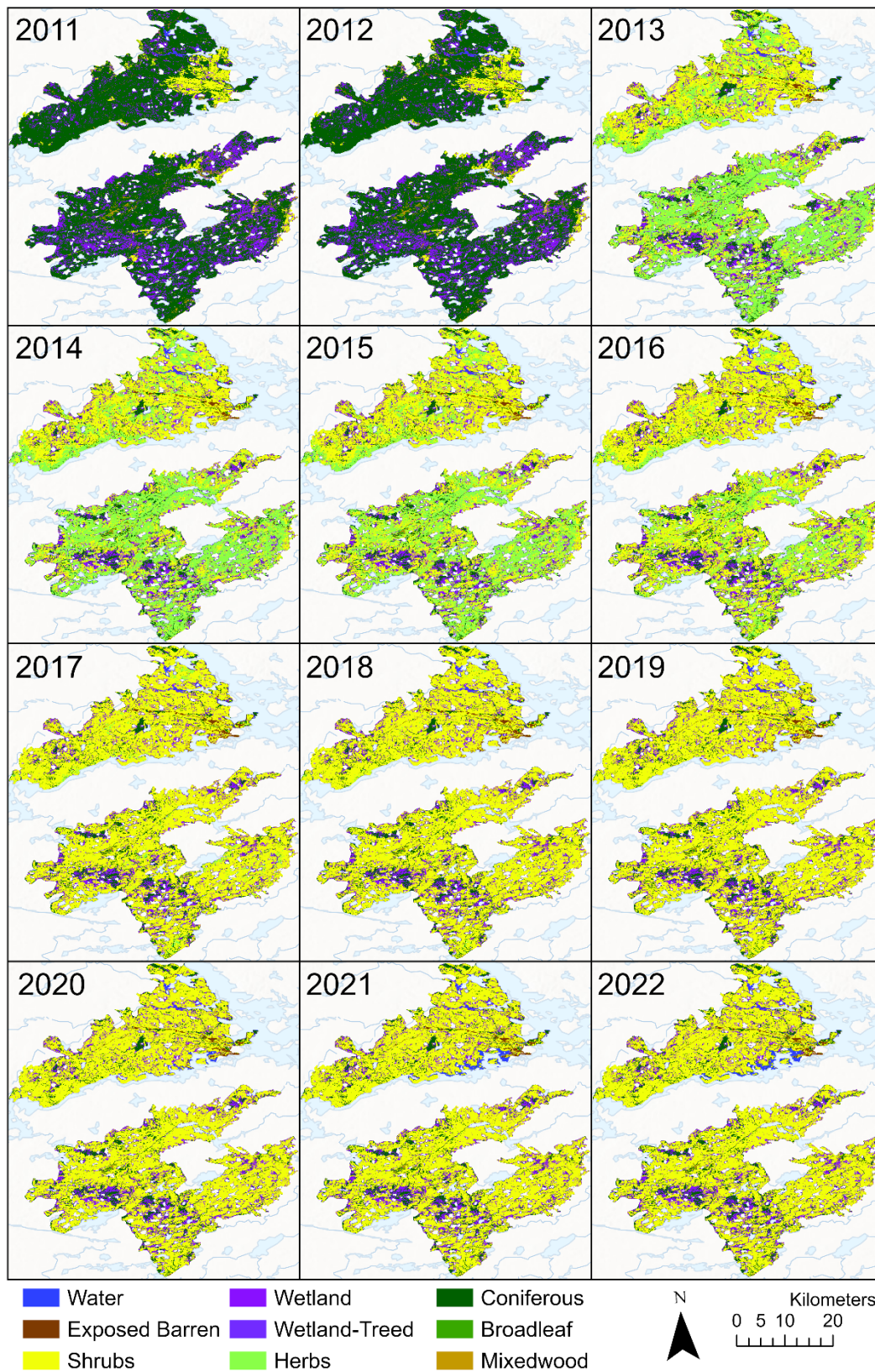


Figure 10. Annual National Terrestrial Ecosystem Monitoring System (NTEMS) land cover maps within the fire perimeter. Panels show pre-fire conditions (2011-2012), the fire year (2013), and post-fire land cover through 2022.

6.0 Discussion

6.1 Spectral recovery patterns

Recovery patterns in this study should be interpreted in the context of a northern Manitoba Boreal Shield landscape that was dominated by black spruce before the fire and includes extensive wetlands and organic soils. NDVI recovered rapidly, with most pixels reaching 80% of their pre-fire values within four years, indicating a return of greenness. This is consistent with earlier boreal time series studies that found faster spectral recovery for NDVI than for NBR (Pickell et al., 2016). This suggests that rapid NDVI recovery reflects early post-fire revegetation rather than the return of pre-fire forest structure, likely driven by the regrowth of shrubs, herbs, and grasses.

NBR recovered more slowly over the full 10-year observation period, and many pixels remained below the 80% threshold by 2023. Pickell et al. (2016) similarly found that NBR recovers more slowly than NDVI because it is more sensitive to vegetation structure and woody biomass. In this study, that pattern is consistent with post-fire recovery in boreal black spruce systems, where fire severity and the combustion of soil organic layers can influence conifer regeneration and successional pathways (Johnstone et al., 2010) (Figure 6 and Figure 7).

Land cover trends derived from NTEMS support this interpretation. After the fire, coniferous forest cover within the fire perimeter declined sharply and remained well below pre-fire levels through the post-fire period, while shrub and herb classes dominated the landscape (Figure 9 and Figure 10). Together, these patterns suggest widespread revegetation rather than the return of pre-fire forest structure. NDVI and NBR reflect different aspects of post-fire recovery: NDVI captures rapid increases in vegetation greenness, while NBR is more sensitive to slower changes associated with vegetation structure and woody biomass.

6.2 Influence of burn severity on spectral recovery

Recovery slopes differed among burn severity classes defined using dNBR. Recovery slopes for NDVI and NBR increased with burn severity from very low or unburned areas to moderate severity classes. Moderate severity pixels showed the highest slopes across land cover types. These patterns suggest that areas with greater initial fire effects also experienced greater spectral change over the following decade (Figure 8).

This pattern does not necessarily indicate stronger ecological recovery in moderate severity areas. Instead, higher recovery slopes likely reflect a larger difference between pre- and post-fire spectral conditions. This created greater potential for spectral values to increase through time. In contrast, very low or unburned areas experienced relatively little spectral change during the fire. These areas therefore showed lower recovery slopes. Because their spectral values remained closer to pre-fire conditions, there was less measurable recovery through time.

6.3 Supporting influence of pre-fire vegetation context

Pre-fire vegetation provides additional context for interpreting the recovery patterns observed in this study. Prior to the fire, the area within the fire perimeter was largely classified as coniferous forest and wetland by the NTEMS land cover product. After the fire, land cover within the fire perimeter shifted first toward herb classes and later toward shrub classes (Figure 9 and Figure 10). These land cover patterns are consistent with the spectral recovery results, with herb and shrub classes dominating the burn area during the post-fire period.

Differences in recovery slopes among pre-fire land cover classes were small. Shrub areas showed somewhat higher slopes, while wetland and wetland-treed areas were slightly lower. This may reflect the moisture conditions typical of wetland environments, which can influence fire severity and the combustion of organic soils (Turetsky et al., 2011). Wetland and upland environments can also differ in their post-fire recovery patterns (Whitman et al., 2019), which may contribute to smaller spectral changes through time. These differences were small relative to the overall recovery patterns observed across the burn area.

6.4 Limitations of spectral recovery analysis

Several limitations affect the interpretation of these results. Burn severity was estimated using dNBR from 2014 rather than imagery collected immediately after the fire, as usable post-fire imagery from 2013 was limited by smoke. By 2014, some vegetation regrowth had already begun, which may have reduced dNBR values and led to an underestimation of burn severity. Landsat imagery also introduces uncertainty at 30 m resolution, since individual pixels can contain mixtures of vegetation types, burn severities, and recovery stages.

Spectral recovery does not equal ecological recovery. A pixel reaching 80% of its pre-fire NDVI or NBR value indicates that certain spectral properties have returned, but it does not confirm the return of pre-fire species composition, canopy height, or ecosystem function. Frazier et al. (2015) identified a data gap in the northern unmanaged zone of the Boreal, where large portions of the ecozone lacked detailed forest information. Without field observations or more detailed data, the ecological meaning of these spectral recovery patterns remains uncertain.

This study did not examine climate variables such as temperature or precipitation. Post-fire climate is an important driver of regeneration outcomes and forest resilience following fire (Stevens-Rumann et al., 2018), and northern Manitoba experiences considerable climate variability. This analysis also covered only the first ten years after the fire, capturing early recovery but not longer-term post-fire change. This is especially important in conifer-dominated systems, where forest structure develops over long time scales following fire (Johnstone et al., 2010).

6.5 Broader implications for boreal wildfire recovery

This study provides local insight into post-fire recovery dynamics in northern Manitoba, where detailed analyses of recovery within individual large wildfires are limited. The results show that

vegetation greenness returned relatively quickly, while other indicators suggest that recovery toward pre-fire forest structure is still ongoing. Continued monitoring will help determine whether the current shrub-dominated conditions transition back toward conifer-dominated forest or persist over longer time scales.

The study area lies within Treaty 5 territory and overlaps with land used by many Indigenous communities. In this context, forest recovery cannot be understood through spectral patterns alone. Christianson et al. (2022) describe how Indigenous peoples in the boreal understand fire as part of a living, interconnected landscape. Spatial information on burn severity, recovery, and land-cover change may still provide one source of information for future land stewardship.

Future work could combine Landsat time series with NTEMS products, field data, or structural information such as canopy height models to better interpret spectral recovery in northern Manitoba's boreal forest landscapes.

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